

# Crowe Sense: the sensing node, and the raise that puts it in real rooms

Prepared 2026-09-01  
For a list of 30 investors and funders  
Every figure measured that day or marked as a proposal

|                                                                                                                |                                                                                                            |                                                                            |                                                                                                     |
|----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| <div>6</div> <div>surfaces reading one contract: desktop, web, phone, Cortex, terminal, any screen</div>       | <div>105</div> <div>tests across firmware, relay, CLI, desktop and Cortex integrations, all green</div>    | <div>\$278</div> <div>parts per node at 4 GB, priced live 2026-08-31</div> | <div>43</div> <div>condition bands from the founder's 68-video corpus, each cited to its clip</div> |
| <div>0</div> <div>nodes that have run with physical sensors in a room. That is the gap this raise closes</div> | <div>\$175<sub>k</sub></div> <div>the ask, on a SAFE; cloud and analytics already covered by credits</div> |                                                                            |                                                                                                     |

Sources: firmware 51, relay 7, CLI 25, desktop 14, Cortex 8 tests; hardware/SHOPPING-LIST.md; analysis/practice-envelope; docs/investor-report.md in [github.com/MichaelCrowe11/crowe-sense](https://github.com/MichaelCrowe11/crowe-sense), branch integrate/2026-09-01.

IN ONE PARAGRAPH

Crowe Sense is a sensing node for a growing room: a Raspberry Pi 5, four Sensirion and Bosch sensors under a louvered radiation shield, a 3D-printed housing, about \$280 in parts. It keeps every reading on the node, serves them on the local network, and pushes signed batches to a relay so the same numbers appear on six surfaces the company already ships. Readings are judged against 43 condition bands extracted from the founder's own cultivation corpus, each cited to the clip it came from.

The software is built and tested. No node has yet run with physical sensors in a room. This raise puts twenty-five of them in ten growers' rooms and proves the data with the same test that caught the company's own synthetic stream in August.

**The node**

Pi 5 · SCD41 · SHT45 · BME688 · VEML7700  
SQLite on the node · local API :8078  
Ed25519-signed batches out



**The relay**

one Cloudflare Worker  
D1 hot 30 days · R2 raw forever  
Crowe ID on every read



**Six surfaces, one contract**

desktop

web

phone + Mac

Cortex

terminal

House / kiosk

contracts/telemetry-v1.md. The local API and the relay serve identical paths, so a surface does not know or care which one it is reading.

WHAT EXISTS TODAY

Measured on 2026-09-01

| PIECE                                                              | STATE            | EVIDENCE             |
|--------------------------------------------------------------------|------------------|----------------------|
| Firmware 0.2.0: sampler, signed uploader, local API, watchdog, six | <div>built</div> | 51 tests, ruff clean |

|                                                                                     |                          |                                                     |
|-------------------------------------------------------------------------------------|--------------------------|-----------------------------------------------------|
| drivers                                                                             |                          |                                                     |
| Relay: signed ingest, D1, R2, Crowe ID reads                                        | built, not deployed      | 7 end-to-end tests including signature verification |
| Contract every surface shares                                                       | written                  | contracts/telemetry-v1.md, JSON schema, D1 schema   |
| Desktop app: the Environment lane fills itself with measured rows                   | branch                   | 14 tests + 67 parity tests                          |
| Terminal: crowe sense, seven subcommands                                            | branch                   | 25 tests; full suite 2,215 green                    |
| Cortex panel                                                                        | branch                   | 8 tests + 28 adjacent; typecheck and build clean    |
| Web edge, /app/sense/*                                                              | patched, awaiting deploy | syntax-checked; one-line deploy                     |
| Phone and Mac client 0.2.0, direct or cloud source                                  | compiles                 | cargo check clean                                   |
| House: embeddable panel and kiosk autostart                                         | built                    | one tag mounts the whole surface                    |
| Enclosures: indoor radiation-shield head; field variant with drive and hotspot bays | parametric OpenSCAD      | print settings and tolerances documented            |
| Practice envelope, provenance test, stage and harvest logs                          | built                    | 23 regression tests, one per mistake it made        |
| Node bill of materials                                                              | priced live              | \$278 at 4 GB, \$343 at 8 GB                        |

NOT PROVEN, SAID FIRST

## What is not proven

**No Crowe Sense node has run with physical sensors in a growing room.** An earlier daemon ran the whole software path for 23 days in May and June 2026 and produced 3,596,420 rows across 20 metrics. On 2026-08-04 the founder's own provenance test

showed the tent temperature, humidity and light channels were not sensor measurements; no sensors had been installed in those tents. The founder confirmed it the same day and drafted a correction to the published Zenodo record (10.5281/zenodo.20722953). The correction is written and goes up before this report goes out.

**No revenue on this product.** The company's software surfaces have no paying subscribers. The founder's businesses have sold cultivation products and courses; none of that is attributed here.

**Yield prediction is not built.** The dataset that would train it, sensor windows joined to harvest outcomes, has zero labelled rows. The tools to log stages and harvests exist; the pilot fills them.

The reason to say this first: the provenance test is the asset. A company that caught its own synthetic data, published the method, and gates every future dataset on it is selling something a sensor vendor cannot: numbers a grower can trust.

THE FOUNDER

Why this founder

Michael Crowe has grown mushrooms since 2005, starting at fifteen, about ten of those years commercially (Southwest Mushrooms, Phoenix; the farm closed in February 2025). About 195,000 people subscribe to his cultivation channel; 68 of those videos were parsed into 642 numeric constants, of which 43 survive as condition bands a sensor can be judged against, with a documented rule that a band needs two independent statements before it judges anything. Nobody else can ship that layer, because nobody else has the corpus. He also built every surface in the table above.

THE DEVICE

The device, in one table

|         |                                                                                                                                              |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Compute | Raspberry Pi 5, 4 GB, active cooler, 64 GB microSD                                                                                           |
| Sensors | SCD41 (NDIR CO2, T, RH), SHT45 (reference T, RH), BME688 (VOC, pressure), VEML7700 (lux); SDP810 differential pressure on the flow-hood node |
| Housing | 3D-printed PETG, louvered radiation-shield head, about 350 g, 14 h print                                                                     |

|                |                                                                                                       |
|----------------|-------------------------------------------------------------------------------------------------------|
| On the node    | SQLite (WAL), local API on :8078 with a kiosk page, Ed25519-signed uploads, nothing deleted when sent |
| Cloud          | one Cloudflare Worker: D1 (30 days hot), R2 (raw batches forever), Crowe ID on every read             |
| Parts cost     | \$278 (4 GB) · \$343 (8 GB)                                                                           |
| Proposed price | \$449 per node, relay included with any paid Crowe ID tier; set after the pilot                       |

THE ASK

\$175,000 on a SAFE

\$175,000

SAFE

Hardware and people only. Cloud, analytics and model inference run on credits the company already holds: Cloudflare \$99,657 to June 2027 (under 1% used), Azure \$20,000, PostHog \$50,000, and \$13,500 in other unclaimed credits. A \$60,000 first tranche (angel or grant) funds the hardware, the engineering contract, the instruments and half the pilot program, and reaches the day-90 milestone.

| USE                                                                                                            | AMOUNT   |
|----------------------------------------------------------------------------------------------------------------|----------|
| Pilot hardware: 25 nodes, printer, reference instruments, spares                                               | \$12,000 |
| Hardware engineering, contract: sensor-head thermal validation, calibration jig, enclosure DFM, pre-compliance | \$35,000 |
| Founder, twelve months                                                                                         | \$60,000 |
| Pilot program: ten growers, installs, travel, stipends                                                         | \$15,000 |
| Compliance and safety review, product insurance                                                                | \$10,000 |
| Legal: SAFE, IP assignment, provisional filing on the envelope method                                          | \$8,000  |
| Calibration and reference instruments                                                                          | \$4,000  |
|                                                                                                                |          |

|                                   |                  |
|-----------------------------------|------------------|
| Contingency (18%)                 | \$31,000         |
| <b>Total</b>                      | <b>\$175,000</b> |
| Cloud, analytics, model inference | \$0              |

#### MILESTONES

### Dated from close, each with its proof

|                    |                                                                                                                                                                                                             |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>day<br/>30</b>  | Five nodes in real rooms at the founder's site and two pilot growers<br>first week of data passes provenance_check.py (exit 0), published with the run log                                                  |
| <b>day<br/>90</b>  | Twenty-five nodes at ten growers; all six surfaces in weekly use<br>relay shows 25 nodes reporting on 6 of 7 days; PostHog funnel, paired to day-7 retention, above 80%                                     |
| <b>day<br/>180</b> | Twenty labelled harvests; Western SARE application submitted (due 2026-10-28, \$35,000, non-dilutive)<br>datasets/outcomes.csv row count; SARE confirmation                                                 |
| <b>day<br/>365</b> | One hundred nodes shipped at \$449; the supervised yield model trained on real labels, or a published reason it cannot be yet<br>shipping records; a model card with held-out error, or the negative result |

#### UNIT ECONOMICS

### Per node, proposed

|                               |                                                        |
|-------------------------------|--------------------------------------------------------|
| Parts                         | \$278                                                  |
| Print, assembly, test (1.5 h) | \$40                                                   |
| Landed cost                   | \$318                                                  |
| Price                         | \$449                                                  |
| Gross margin                  | 29% at pilot volume; parts fall about 20% at 100 units |

|            |                                                                 |
|------------|-----------------------------------------------------------------|
| Cloud cost | \$0 through June 2027 on credit; under \$1 per node-month after |
|------------|-----------------------------------------------------------------|

RISKS

# Risks, plainly

| RISK                                                      | WHAT IS DONE ABOUT IT                                                                                                                                         |
|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pi 5 pricing (the 2026 DRAM shortage doubled 2024 prices) | 4 GB board; the firmware runs on a Pi 4 or a Compute Module if it comes to that                                                                               |
| Sensor drift, especially NDIR CO2                         | SHT45 as reference, SCD41 self-calibration on, every reading carries its driver and a quality flag; reference instruments in the budget                       |
| A second provenance failure                               | the test runs on every node's first week and is published with the data                                                                                       |
| Founder dependence                                        | every surface, the contract and the setup pipeline are documented in the repo; a contractor built the desktop and Cortex integrations from the contract alone |
| Market size                                               | not claimed here; the pilot measures whether ten growers keep the nodes plugged in                                                                            |

THE THIRTY

# Thirty investors and funders

Ten agtech and controlled-environment funds, six hardware and climate seed funds and accelerators, eight Arizona programs and angel groups, six non-dilutive programs. Every URL was verified active in 2025 or 2026 on 2026-09-01; award ceilings are the current published figures. Contact names are deliberately absent: they change, and a wrong one costs more than a blank. Start with HAX, Techstars Future of Food, Canyon Angels, Desert Angels, Arizona Tech Investors and the USDA and NSF SBIR windows; the large agtech funds are later-round relationships.

# Agtech and controlled-environment funds 10

| NAME | STAGE OR AWARD | WHY IT FITS |
|------|----------------|-------------|
|------|----------------|-------------|

|                              |                                                                                       |                                                                                                                                                                                                                   |
|------------------------------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>AgFunder</b>              | Seed to Series B, broad agrifood and climate tech                                     | AgFunder tracks agrifood infrastructure deals and has backed sensor and data plays comparable to a Raspberry Pi based environmental node for indoor mushroom and CEA growers.                                     |
| <b>Anterra Capital</b>       | \$1,000,000 to \$10,000,000 initial check (stated on the fund's own site)             | Anterra's thesis centers on digital and data infrastructure across the agrifood value chain, matching the sensor-plus-Cloudflare-relay architecture behind Crowe Sense.                                           |
| <b>Astanor Ventures</b>      | Seed through growth                                                                   | Astanor explicitly funds controlled environment agriculture and impact-native agrifood founders, a direct thesis match for an indoor-growing sensing platform.                                                    |
| <b>iSelect Fund</b>          | Seed, Series A, crop health and sensing                                               | iSelect's stated mission of funding sensor-based farm management tools and its indoor-farming portfolio company Artemis show current interest in exactly the CO2/VOC/climate sensing hardware Crowe Sense builds. |
| <b>Ospraie Ag Science</b>    | Seed through growth, controlled environment agriculture named as a backed theme       | Ospraie explicitly names controlled environment agriculture as a backed theme, making Crowe Sense's CEA sensing node a direct thesis match.                                                                       |
| <b>Middleland Capital</b>    | Seed, Series A (recent language skews toward de-risked, early-revenue companies)      | Middleland's portfolio already includes precision-ag sensing hardware (EarthOptics), showing comfort underwriting physical agricultural sensing plays like Crowe Sense.                                           |
| <b>Acre Venture Partners</b> | Seed through Series B                                                                 | Acre's current fund already holds an indoor-farming software company (Source) and ag-robotics hardware bets, placing a hardware-plus-software CEA sensing platform inside their thesis.                           |
| <b>Better Food Ventures</b>  | Seed, Series A, digital/software-first (deliberately avoids hardware-intensive deals) | Better Food Ventures backs the software side of food and ag systems, so a pitch should lead with the Cloudflare relay, apps and CLI rather than the physical sensor hardware.                                     |
|                              |                                                                                       |                                                                                                                                                                                                                   |

|                                |                                                                              |                                                                                                                                                                                                           |
|--------------------------------|------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Radicle Growth</b>          | verify                                                                       | Radicle's challenge-based sourcing model (2025 AgRogue Growth Partners with Land O'Lakes) is built to surface sensor-and-data farm technology like Crowe Sense.                                           |
| <b>SVG Ventures<br/>THRIVE</b> | Up to \$1,000,000 for Global Impact Challenge finalists (stated by the fund) | THRIVE's recent portfolio includes storage-monitoring sensor company Cellar Insights (Jan 2026), and its Global Impact Challenge funds early agrifood hardware and software at Crowe Sense's exact stage. |

## Hardware and climate seed funds and accelerators 6

| NAME                                              | STAGE OR AWARD                                                                                                                                                               | WHY IT FITS                                                                                                                                                                                                      |
|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Lemnos</b>                                     | Pre-seed and seed, hardware and deep tech only                                                                                                                               | Lemnos is a hardware-exclusive early-stage fund with a track record in IoT sensing hardware, matching a Raspberry Pi based environmental node moving from firmware-complete into pilot manufacturing.            |
| <b>HAX (SOSV)</b>                                 | Up to \$550,000 pre-seed; first check \$250,000 (\$150,000 cash plus \$100,000 in-kind engineering and lab support) on an uncapped post-money SAFE (stated on HAX's own FAQ) | HAX is a pre-seed program with in-house mechanical/electrical engineers and small-batch manufacturing support, directly addressing the enclosure, sensor integration and pilot production run Crowe Sense needs. |
| <b>Lowercarbon Capital</b>                        | Pre-seed through growth, climate tech with a hardware-heavy sub-portfolio                                                                                                    | Lowercarbon funds deep-tech climate hardware and has in-house engineering diligence capacity that could evaluate a CO2/VOC/differential-pressure sensing node as CEA climate infrastructure.                     |
| <b>Techstars Future of Food Powered by Ecolab</b> | \$220,000 (Techstars standard terms: \$20,000 for 5% common plus \$200,000 uncapped MFN SAFE)                                                                                | Successor to Techstars Farm to Fork with an active 2026 cohort; the Cloudflare relay and multi-platform apps behind Crowe Sense fit its data/traceability angle better than the sensor hardware alone.           |
| <b>Y Combinator</b>                               | \$500,000 (\$125,000 for 7% plus \$375,000 uncapped MFN SAFE)                                                                                                                | YC's 2026 Requests for Startups name agriculture robots and physical hardware as priority categories, and its flat \$500k check would fund a Crowe Sense pilot production run outright.                          |

|                         |                                                                                                              |                                                                                                                                                                                                  |
|-------------------------|--------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Elemental Impact</b> | First-of-a-kind and early commercial deployment funding across energy, industry, food and agriculture, water | Elemental Impact funds food-and-agriculture climate hardware moving from prototype into real-world deployment, matching Crowe Sense's move from unit-tested firmware into a validated pilot run. |
|-------------------------|--------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Arizona 8

| NAME                                      | STAGE OR AWARD                                                                              | WHY IT FITS                                                                                                                                                                                                    |
|-------------------------------------------|---------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Desert Angels</b>                      | Seed/early-stage, accredited angel network                                                  | Arizona's top-ranked angel network per the HALO Report gives a Phoenix hardware startup access to the state's deepest bench of accredited investors and sidecar funds.                                         |
| <b>Canyon Angels</b>                      | Pre-seed/seed, accredited angel group                                                       | A Phoenix angel group meeting on the Grand Canyon University campus is a realistic first local stop for a Phoenix-based hardware founder with an existing regional audience.                                   |
| <b>Invest Southwest (Venture Madness)</b> | not applicable (event, not a fund)                                                          | Arizona's longest-running venture conference puts a Phoenix hardware/agtech founder directly in front of the state's active accredited investors during Arizona Tech Week.                                     |
| <b>Tallwave Capital</b>                   | Pre-seed, seed and post-seed VC fund                                                        | Named Arizona's most active venture capital firm two years running, Tallwave gives a Phoenix hardware startup a credible local lead investor plus a structured validation track through its High Tide program. |
| <b>Arizona Tech Investors (ATI)</b>       | verify                                                                                      | Arizona's largest angel organization lists advanced manufacturing and sustainability among its target sectors, a direct route to reach 100+ accredited local angels.                                           |
| <b>PHX Ventures</b>                       | \$50,000 to \$250,000 in pre-revenue companies (Launch Fund, stated on the fund's own site) | The Launch Fund's stated stage and check-size range matches Crowe Logic almost exactly, though the fund states it invests exclusively in B2B software and AI, not hardware.                                    |
|                                           |                                                                                             |                                                                                                                                                                                                                |

|                           |                                                                                                          |                                                                                                                                                                                                    |
|---------------------------|----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>AZ TechCelerator</b>   | not applicable (space and mentorship, not capital)                                                       | A hardware-oriented Arizona incubator offers affordable space and mentors useful for pushing a sensor enclosure and pilot run through validation, though it is not a source of investment capital. |
| <b>ASU Venture Devils</b> | \$1,000 to \$50,000 per team; Spring 2026 Demo Day pool exceeded \$300,000 in aggregate (ASU's own site) | ASU's venture competition offers non-dilutive prize money at Crowe Logic's exact stage, but eligibility requires an enrolled ASU or Maricopa Community College student as the team's lead founder. |

## Non-dilutive programs 6

| NAME                                                             | STAGE OR AWARD                                                                                          | WHY IT FITS                                                                                                                                                                                                        |
|------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>USDA SBIR/STTR Phase I (NIFA)</b>                             | \$175,000 maximum for most topics (\$125,000 for topics 8.6 and 8.12), award ceiling not guaranteed     | Funds exactly the CO2/VOC/climate sensor hardware and firmware R&D Crowe Sense needs to validate for indoor mushroom and CEA cultivation.                                                                          |
| <b>NSF SBIR Phase I (America's Seed Fund)</b>                    | \$305,000 maximum (solicitation NSF 26-510), award ceiling not guaranteed                               | NSF explicitly funds small-business deep-tech hardware R&D, and a multi-sensor Pi-based node with a Cloudflare relay is exactly that kind of hardware-plus-software integration project.                           |
| <b>Western SARE Farmer/Rancher Grant</b>                         | \$35,000 flat award                                                                                     | Michael Crowe, as an active mushroom producer, can serve as producer-PI piloting Crowe Sense units on his own grow as a funded on-farm research and outreach project.                                              |
| <b>USDA NIFA AFRI, Agriculture Systems and Technology</b>        | verify (no per-track cap published; check the current NOFO)                                             | This AFRI priority area explicitly covers precision-agriculture sensing and engineering, matching Crowe Sense's environmental sensing platform for CEA more closely than AFRI's larger systems-level tracks.       |
| <b>Arizona Innovation Challenge (Arizona Commerce Authority)</b> | Minimum \$50,000 non-dilutive per awardee, up to 10 awardees per cycle (official ACA July 2025 release) | A state-run non-dilutive competition whose eligible industries include hardware and advanced manufacturing fits a pre-revenue Phoenix hardware company seeking runway toward a pilot run without giving up equity. |
|                                                                  |                                                                                                         |                                                                                                                                                                                                                    |

|                                                                    |                                                                                                                                  |                                                                                                                                                                                                          |
|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Arizona Specialty Crop<br>Block Grant Program<br>(AZDA / USDA AMS) | verify (approximately<br>\$1,250,000 total FY2026<br>Arizona allocation across all<br>awardees, no per-project cap<br>published) | Cultivated mushrooms are a USDA-<br>recognized specialty crop, so a sensing<br>project framed as a broad benefit to Arizona<br>mushroom and CEA growers, not a single<br>company's product, can qualify. |
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Crowe Logic, Inc. · [github.com/MichaelCrowe11/crowe-sense](https://github.com/MichaelCrowe11/crowe-sense), branch `integrate/2026-09-01`: firmware, relay, contract, enclosures, archive with the correction, analysis, the six integrations, the shopping list, the setup pipeline, and this report as `docs/investor-report.md`. Prepared 2026-09-01. Nothing here is a projection dressed as a fact.